CART 498

November 5, 2026

# OLLAMA

*Making AI accessible, localized, and private*

**Presentation By:**

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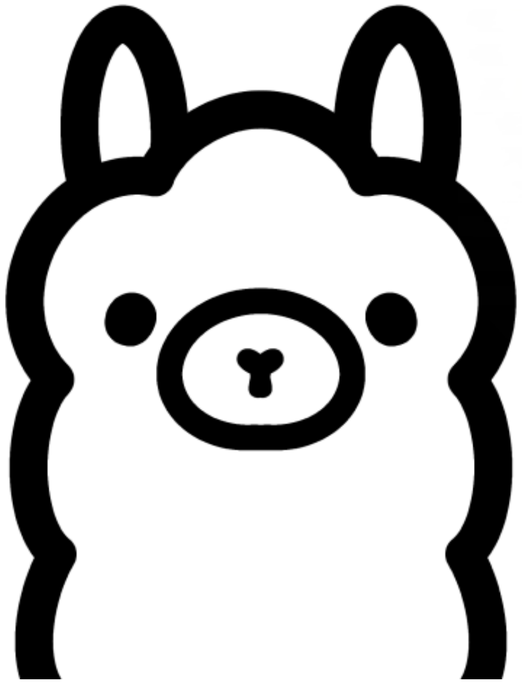
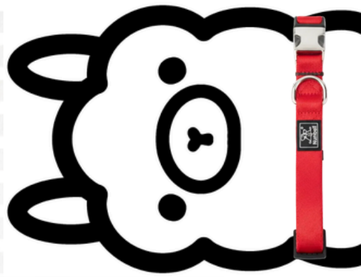
# INTRODUCTION

Ollama is an open-source application that allows you to **run Open-Source AI models on your local machine.**

It uniquely provides an **affordable, streamlined, private access point** to open AI models. You decide whether to strictly use **your processing power or the private Ollama cloud**. (Ollama 2026)



# IMPACT



## BEFORE OLLAMA:

*Incompatibilities / Access Issues*

- Open Source AIs were decentralized and had different requirements for use. (Open Source Initiative n.d.)

*GPU Cost*

- Many larger open source models were unavailable to simple desktop computers because of memory demand.

## AFTER OLLAMA

*Access Simplified*

- Ollama allows users to download & run open source AIs without an API key or expensive hardware for larger open AI models.

*Affordability*

- Free to use & the Ollama cloud is cheaper than most cloud uses. (Ollama 2025)

*Privacy*

- Trusted, secure, private cloud services provided

# MAIN FEATURES

## PRIVACY

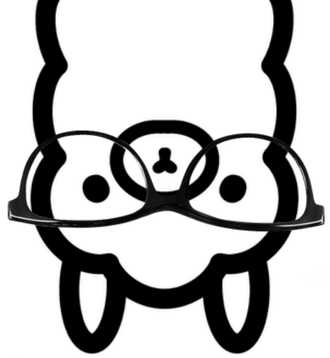
- Ollama keeps your prompts & data locally on your device; even on the Ollama cloud, **your usage information is never stored remotely or used to train models**. (Ollama 2026)

## COST & AFFORDABILITY

- Using models locally with Ollama is free because it uses your GPU/CPU power. Paid tokens get you $60 of credit for $20. (Ollama n.d.)

## OFFLINE USAGE

- Whether you’re out in public or stuck with no internet connection, it is still possible to use AI models with Ollama as long as you have downloaded a model. Also allows developers to host their own models and charge for their own GPU usage. (Ollama 2025)





# USE CASES

## CHAT INTERFACES

**Web:** web-based Ollama Cloud hosted
**Desktop:** downloadable & local specific AIs
**Mobile:** access to Open Source AIs through your phone, run on the Ollama Cloud.

## PRODUCTIVITY APPS

Hosted on GitHub, developers post creative AI-powered Chrome extensions, downloadable painting apps, or a hybrid of locally hosted web assistants.

## OBSERVABILITY & MONITORING

Devs make downloadable apps to assist workflows while storing all the information on your device.

## COMMUNITY INTEGRATIONS

Ollama github (github.com/ollama/ollama) supports pushing your own applications to the README.md or remixing existing projects

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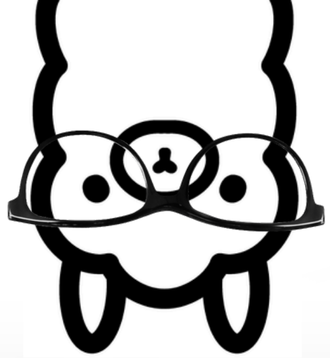

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# USE CASE #1

## CHAT INTERFACES

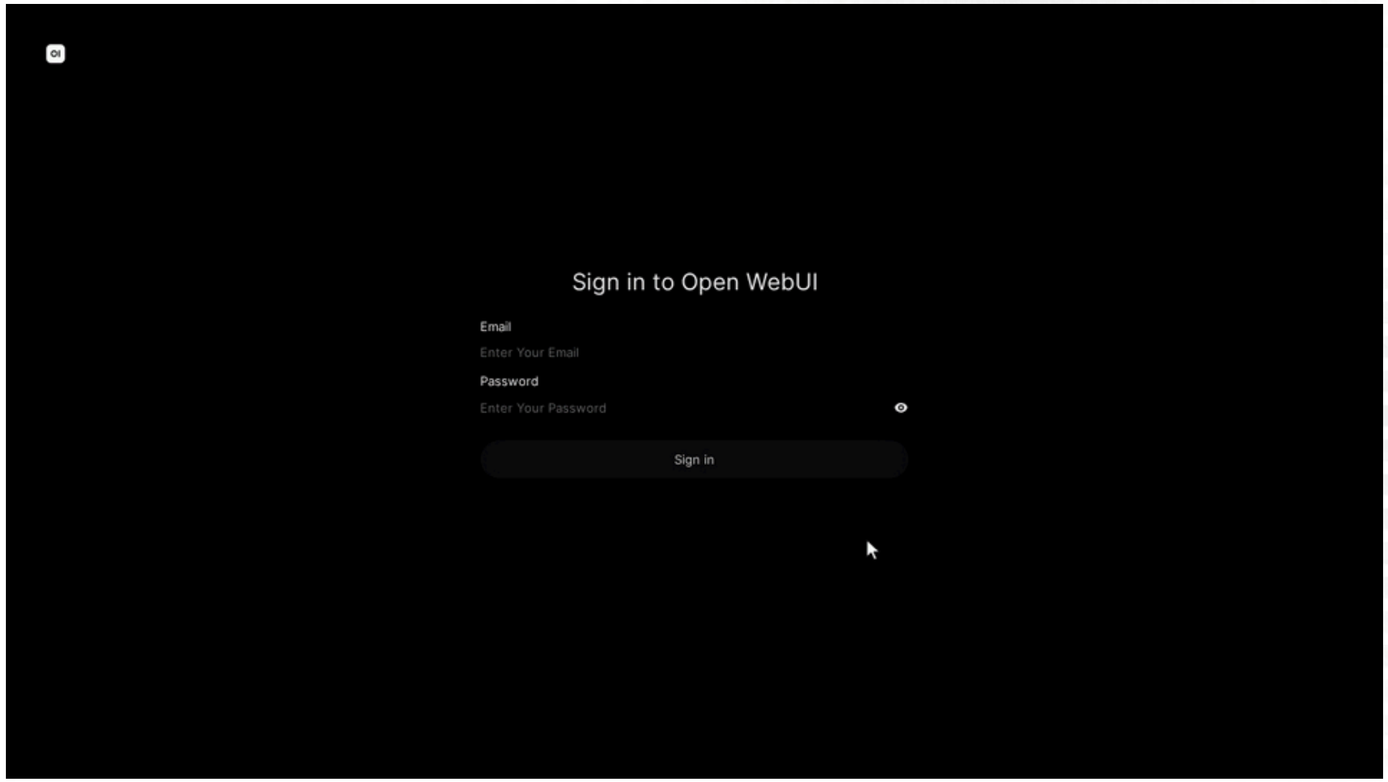
**Web:** web-based Ollama Cloud hosted
**Desktop:** downloadable & local specific AIs
**Mobile:** access to Open Source AIs through your phone, run on the Ollama Cloud.

## EXAMPLE:

**Open Web UI**

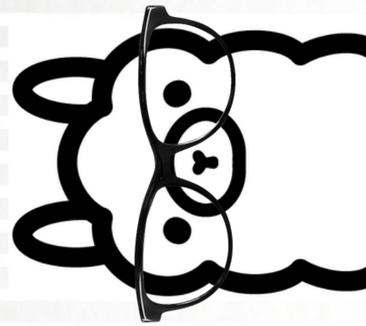
- Open Web UI Is a platform built enturely to run offline, offered in multuple languages (open-webui n.d.)

English | 简体中文 | 繁體中文 | 한국어 | Deutsch | Español | Français | Italiano | Dansk | 日本語 | Polski | Русский | Bosanski | العربية | Norsk | Português (Brasil) | ไทย | Türkçe | Українська | বাংলা | Ελληνικά | Tiếng Việt




https://github.com/open-webui/open-webui

# USE CASE #2

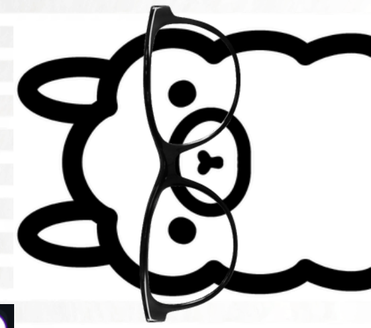


## PRODUCTIVITY

**Interviewr - ai** is an open source application that assists you with job interviews by helping you practice and providing assistance during the interview Itself. Download onto your device via github. (PSYEONE n.d.)

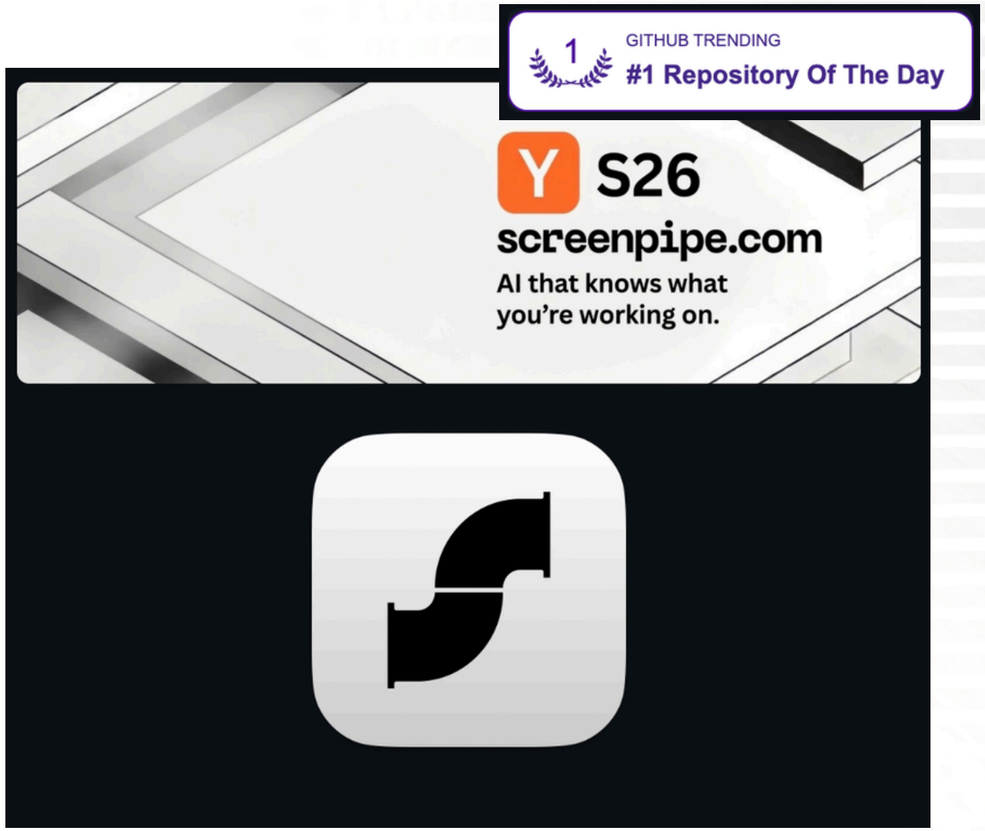
https://github.com/PSYEONE/interviewr-ai

# USE CASE #3



## OBSERVABILITY & MONITORING

1. **screenpipe** continuously records your company's computer work, maps workflows, and helps find work worth automating. Run locally so this information never leaves your device. (screenpipe n.d.)




https://github.com/screenpipe/screenpipe

# DOWNLOADING OLLAMA

Downloading Ollama for everyday use is quick and straightforward, just **open your computer terminal and paste run command:**

**macOS**

```
curl -fsSL https://ollama.com/install.sh | sh
```

**Windows**

```
irm https://ollama.com/install.ps1 | iex
```

**Linux**

```
curl -fsSL https://ollama.com/install.sh | sh
```

and you’re good to go!

# COST

## FREE TIER

- Unlimited local model use
- No usage credits, Server processes up to 3 concurrent requests
- One cloud model at a time
  - Cloud models are larger models too big to run on local software (Ollama n.d.)

## OLLAMA PRO ($20/MONTH)

- $60 of usage credits per month, amount of tokens depends on the models demands. Allows you to run models on ollama's GPUs. No explicit token amount is discplosed.
- 3 cloud models at once (Ollama n.d.)

## OLLAMA MAX ($100/MONTH)

- 5x pro usage,
- 10 cloud model concurrently (Ollama n.d.)

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# HARDWARE

## WINDOWS

VRAM
(GPU USAGE)

## MAC

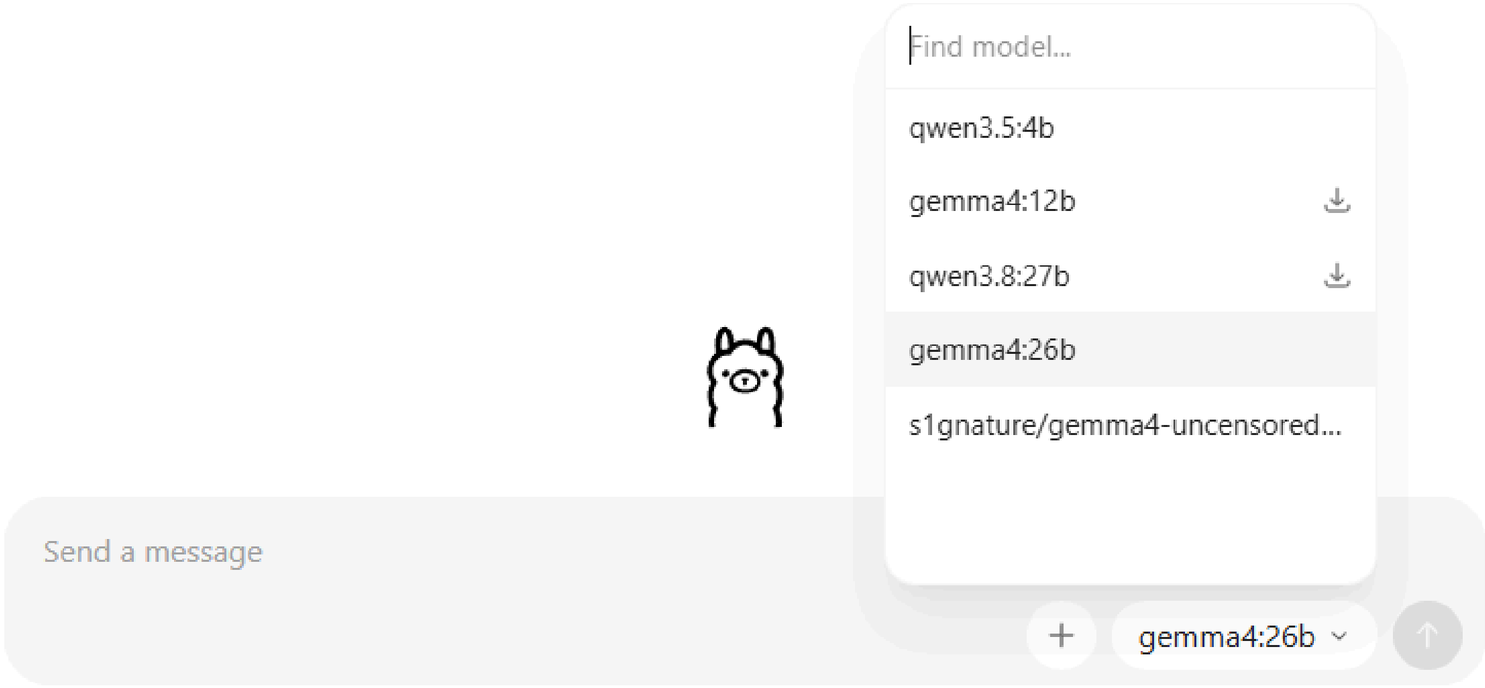
REGULAR
RAM

(SHARED BETWEEN THE PROCESSOR AND THE GPU)

## NO GPU

EXTREMELY
SLOW

*RECAP


Find model...
qwen3.5:4b
gemma4:12b
qwen3.8:27b
gemma4:26b
s1gnature/gemma4-uncensored...
Send a message
gemma4:26b

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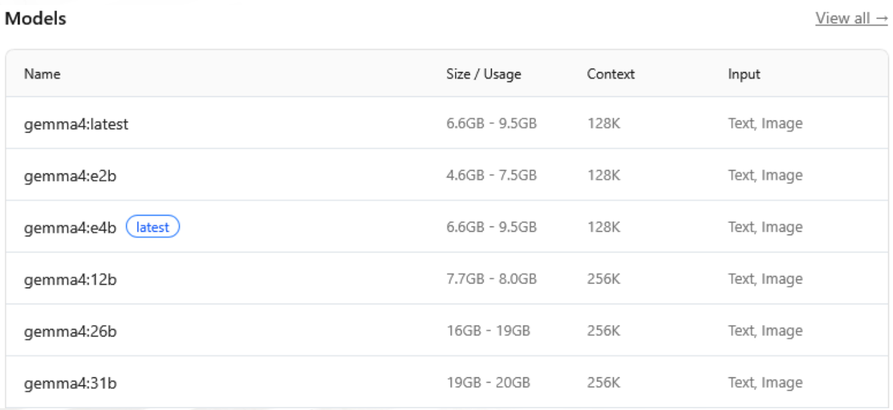
# CAPABILITY TAGS

vision tools thinking cloud

Ollama's website provides capability tags disclosing information on the model's functions

: 2b 4b 9b 27b

It will also disclose the amount of parameters included in the model ('b' for billion)



Models

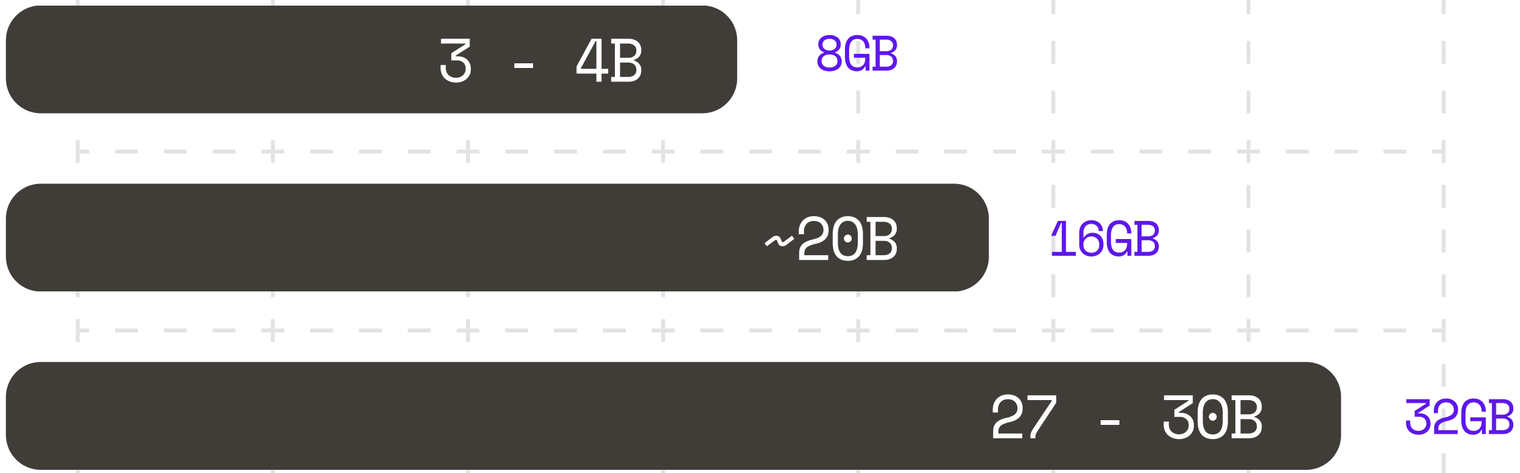
View all →

| Name | Size / Usage | Context | Input |
|---|---|---|---|
| gemma4:latest | 6.6GB - 9.5GB | 128K | Text, Image |
| gemma4:e2b | 4.6GB - 7.5GB | 128K | Text, Image |
| gemma4:e4b latest | 6.6GB - 9.5GB | 128K | Text, Image |
| gemma4:12b | 7.7GB - 8.0GB | 256K | Text, Image |
| gemma4:26b | 16GB - 19GB | 256K | Text, Image |
| gemma4:31b | 19GB - 20GB | 256K | Text, Image |

1M $$$$

token usage for cloud based models will be displayed

# RECOMMENDED (BASED ON VRAM)



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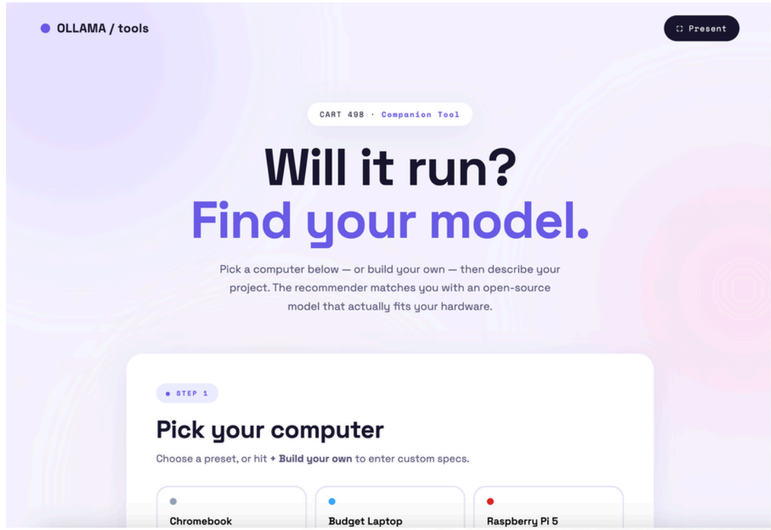
# HARDWARE REQUIREMENTS

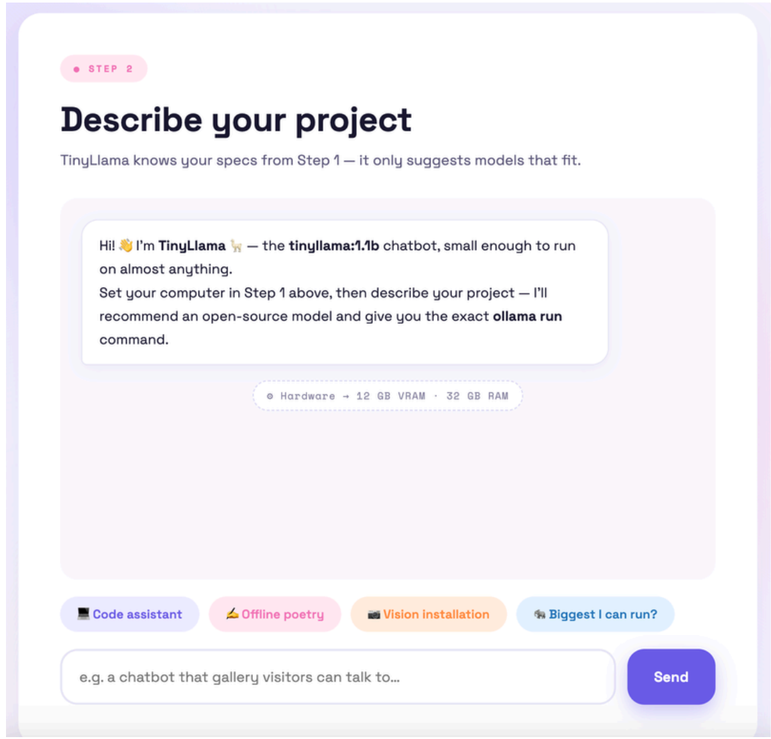
**For one of our creative pieces,** we made a website that helps you determine what model is best for you!

It checks what model you can run locally on your device, and using a custom tinyllama:1.1b chatbot, it will recommend an Open AI for your project.

(Made with kimi-k3:cloud via Ollama)

https://browneyegirl.github.io/CART498-GenAI/A06/findyourmodel_creativepiece_1/



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# INTERACTION PARADIGM

**Human agency**

- Define tasks, instructions, and desired outputs.
- Control context and refine inputs.
- Choose and switch between AI models.
- Direct tasks such as summarizing, analyzing, and brainstorming.
- Evaluate and filter AI-generated responses.
- Control data privacy through local processing.
- Run models independently of internet access.

# WHAT DO YOU CONTROL?

- INSTRUCTIONS AND CONTEXT
- LOCAL OR CLOUD-HOSTED
- MANAGING THE MODELS ON YOUR MACHINE

**Terminal command examples:**

| | |
|---|---|
| *ollama pull <model>* | download model |
| *ollama run <model>* | run model |
| */bye* | exit chat |
| *ollama list* | list commands |
| *ollama rm <model>* | remove model |
| *ollama ps* | display loaded models |

# LIMITATIONS

## Model limitations

- Limited model capabilities
- Invents information
- Usage limits
- Model availability

## Hardware limitations

- Ram & storage
- Mid-size models can’t be pushed to github

## Other

- Ollama still has power over your access
- Cloud prompts saved for 5 minutes after
- Information stored in GPUs back in USA & Europe
- Output biases of open source models it hosts

# FINAL PROJECTS

*IS OLLAMA RIGHT FOR YOU?*

## APPLICATION

- Do you need to use AI in a downloadable application?
  - You can download mini AI models through Ollama
- Do you need an AI that only needs to do specific tasks?
  - Open AI models through Ollama can be more light weight if they only do one specific task.
- Are you on a budget?
  - Ollama allows you to download models for free, they will never cost money. You own the model.

# REFERENCES

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# THANK YOU

**Presentation By:**

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